How Does Language Organize Emotions?

Distributional Semantics Meets Valence, Arousal and Dominance Norms

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Abstract

Do emotion words that are used in similar linguistic contexts also share similar psychological properties? I compare two independent representations of 53 English emotion words: Word2Vec embeddings trained on Google News, which capture how words are used, and the valence–arousal–dominance (VAD) ratings of Warriner et al. (2013), which capture how they are experienced. Pairs of emotions whose embeddings are similar (cosine ≥ 0.50) are markedly closer in VAD space than other pairs (mean Euclidean distance 2.26 vs. 3.63; Welch's $t = -7.58$, $p < 10^{-11}$), a result that survives a permutation test respecting the dependence between pairs ($p < 10^{-4}$). A logistic regression on the three per-axis differences shows that valence and dominance each predict semantic proximity, with effects of almost equal size once standardized, while arousal has no detectable effect. Contrary to my initial hypothesis, valence is therefore not the only dimension that structures emotional vocabulary: the sense of control an emotion conveys matters as much.

1 Introduction

When we think about emotions, we spontaneously group some of them together: *joy* and *happiness* feel related, while *joy* and *anger* feel like opposites. Psychologists have long proposed that this intuition can be described with a small number of affective dimensions. Following the semantic differential of Osgood et al. (1957) and the pleasure–arousal–dominance model of Mehrabian and Russell (1974), three dimensions have become standard: *valence* (negative to positive), *arousal* (calm to excited) and *dominance* (overwhelmed to in control). Warriner et al. (2013) collected such ratings for almost 14,000 English lemmas.

Computational linguistics offers a very different route to word meaning. Following the distributional hypothesis (Harris, 1954), models such as Word2Vec (Mikolov et al., 2013a,b) learn a vector for each word from the contexts in which it appears, without any human labelling. Two words used in similar contexts end up with similar vectors.

These two representations are built from entirely different data: one from introspective judgements, the other from co-occurrence statistics in news text. Comparing them asks whether the way we *use* emotion words already encodes how we *experience* the emotions they name, and if so, along which dimensions. My hypothesis was that valence, the most salient affective dimension (Russell, 1980), would be the main organizing factor, with arousal and dominance playing a minor role.

2 Data and methods

2.1 Emotion words and VAD norms

I selected 53 common English emotion words spanning basic emotions (*anger*, *fear*, *joy*) and more complex ones (*compassion*, *embarrassment*, *gratitude*), together with a few general evaluative

terms (*good*, *bad*). All 53 appear in the Warriner et al. (2013) norms, from which I take the mean valence, arousal and dominance ratings (each on a 1–9 scale). Figure 1 shows their distribution. Valence is strongly bimodal, separating negative from positive emotions, whereas arousal is concentrated in the middle of the scale.

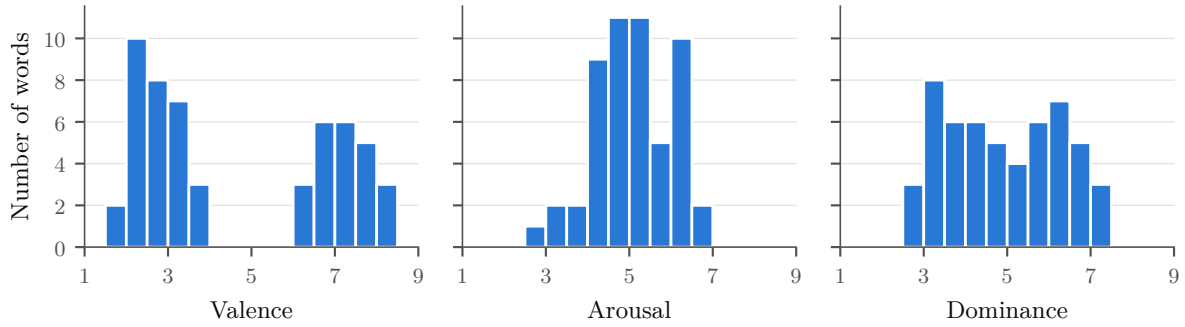


Figure 1: Distribution of the VAD ratings of the 53 emotion words (Warriner et al., 2013).

2.2 Semantic links from Word2Vec

For every pair of words I computed the cosine similarity of their 300-dimensional Word2Vec vectors (Google News model). A pair is considered *semantically linked* if its similarity is at least 0.50. Similarities between related words typically fall between 0.3 and 0.7 in this model, and 0.50 retains clearly related pairs without being overly restrictive. This yields 89 links. Five words have no link at this threshold (*awe*, *contempt*, *relief*, *terror*, *want*) and are excluded, leaving 48 words and $\binom{48}{2} = 1{,}128$ pairs, of which 7.9% are linked.

2.3 Statistical analysis

For each pair (i, j) I compute the absolute differences on each axis, ΔV, ΔA and ΔD, and the Euclidean distance $d_{ij} = \sqrt{\Delta V^2 + \Delta A^2 + \Delta D^2}$ in VAD space. Two questions are then asked.

1. **Are linked pairs closer in VAD space?** I compare the mean distance of linked and unlinked pairs with Welch's t-test. Because each word appears in 47 pairs, pairs are not independent. As a robustness check, I therefore also run a permutation test in the spirit of the quadratic assignment procedure (Krackhardt, 1988): the VAD profiles are randomly reassigned to words 10,000 times while the network of links is held fixed, and the observed difference in mean distance is compared with this null distribution.

2. **Which dimensions matter?** I fit a logistic regression predicting the binary link status from ΔV, ΔA and ΔD. Since the three differences have different spreads (standard deviations of 2.07, 0.78 and 1.12), I also report coefficients for standardized predictors, which make the effects directly comparable.

3 Results

3.1 The semantic network in affective space

Figure 2 projects the 48 words onto the two dimensions that turn out to matter, valence and dominance, and draws the 89 semantic links. Two dense clusters appear: a low-valence, low-dominance cluster (*anxiety*, *fear*, *grief*, *frustration*, *panic*) and a high-valence, high-dominance cluster (*joy*, *happiness*, *optimism*, *contentment*). Most links (70 of 89) connect two words on the same side of the valence midpoint. Many of the long links crossing the valence axis connect

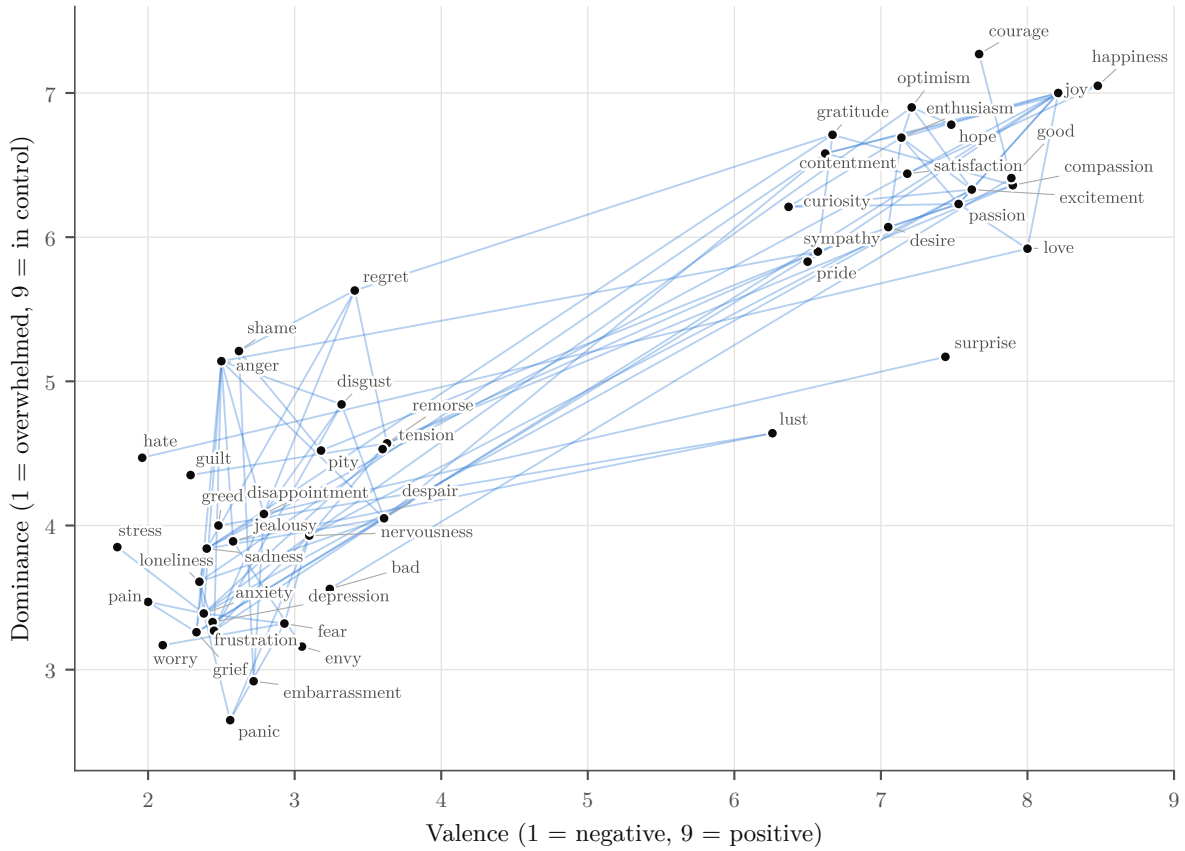


Figure 2: The 48 linked emotion words placed by valence and dominance (Warriner et al., 2013). Lines connect pairs whose Word2Vec cosine similarity is at least 0.50 (89 links). Arousal, the third dimension, is not shown.

antonyms, such as *love–hate*, *good–bad* or *joy–sadness*, which Word2Vec is known to place close together because they appear in the same contexts.

3.2 Linked emotions are psychologically closer

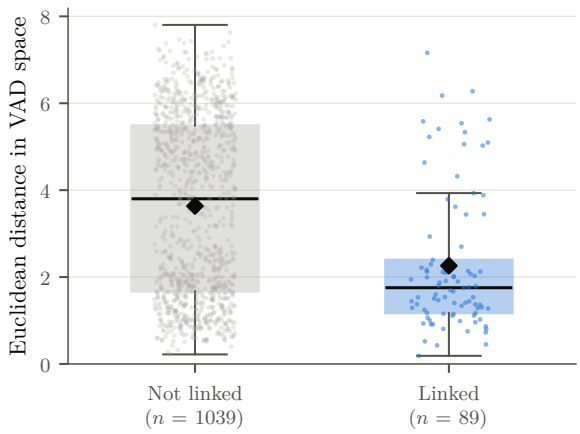
Linked pairs are substantially closer in VAD space than unlinked pairs (Table 1, Figure 3a). The mean distance drops from 3.63 to 2.26 and the median from 3.80 to 1.75, a difference of about 0.7 pooled standard deviations. Welch's test gives $t(114) = -7.58$, $p = 9.9 \times 10^{-12}$. The permutation test confirms the result: none of the 10,000 random reassignments produced a difference as large as the one observed ($p < 10^{-4}$), so the effect is not an artefact of the dependence between pairs.

Pairs	n	Mean distance	SD	Median
Linked	89	2.26	1.60	1.75
Not linked	1,039	3.63	2.04	3.80

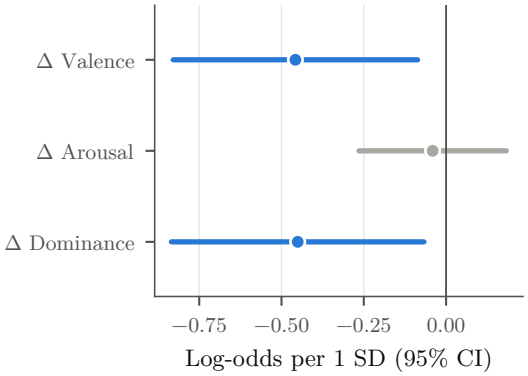
Table 1: Euclidean distance in VAD space by semantic link status. Welch's $t(114) = -7.58$, $p < 10^{-11}$; permutation test (10,000 draws): $p < 10^{-4}$.

3.3 Valence and dominance, not arousal

Table 2 reports the logistic regression. Differences in valence ($p = 0.016$) and dominance ($p = 0.021$) both significantly lower the probability that two emotions are semantically linked, while the difference in arousal has no detectable effect ($p = 0.72$). The model as a whole is highly significant (likelihood-ratio $p = 2.6 \times 10^{-9}$), although its modest pseudo-R^2 (0.07) shows that



(a) VAD distance by link status. Diamonds mark the means.



(b) Standardized logistic regression coefficients.

Figure 3: Main results. Grey marks a non-significant effect.

VAD differences explain only part of the semantic structure.

In raw units, the dominance coefficient (-0.40) is about twice the valence coefficient (-0.22). This comparison is misleading, however, because a one-point difference in dominance is rarer than a one-point difference in valence. With standardized predictors (Figure 3b), the two effects are almost identical: one standard deviation of additional distance reduces the odds of a link by 37% for valence ($\beta = -0.46$) and by 36% for dominance ($\beta = -0.45$). The two predictors are also strongly correlated ($r = 0.75$ between ΔV and ΔD), as positive emotions tend to be felt as controllable. Their separate contributions should therefore be read as roughly equal rather than ranked.

	β	SE	z	p	Odds ratio	β_{std}
Intercept	-1.458	0.230	-6.34	< 0.001	–	–
ΔV (valence)	-0.221	0.091	-2.42	0.016	0.80	-0.458
ΔA (arousal)	-0.052	0.146	-0.36	0.720	0.95	-0.041
ΔD (dominance)	-0.403	0.175	-2.30	0.021	0.67	-0.451
$N = 1{,}128$ pairs; McFadden pseudo-$R^2 = 0.069$; LR test $p = 2.6 \times 10^{-9}$						

Table 2: Logistic regression of semantic link status on absolute VAD differences. Odds ratios are per one-point difference on the 1–9 scale; β_{std} is the coefficient per standard deviation.

4 Discussion

The way emotion words are distributed in news text does reflect how people rate the corresponding emotions. Emotions that are used in similar contexts are, on average, close in affective space, and this proximity is carried by two dimensions. Valence is one of them, as expected. The surprise is dominance: whether an emotion makes us feel in control (*pride*, *courage*) or overwhelmed (*panic*, *embarrassment*) shapes its linguistic neighbourhood as much as whether it is pleasant. My initial hypothesis that valence alone organizes emotional vocabulary is therefore only partly supported.

The absence of an arousal effect is also informative. Emotions of very different intensity are often linked: *grief* and *sadness*, *happiness* and *satisfaction*, or *enthusiasm* and *excitement* are close in usage despite arousal gaps of two to three points. One interpretation is that language groups emotions by *what kind* of situation they evaluate rather than by *how strongly* they are felt, intensity being expressed by other means (adverbs, intensifiers, word choice within a family).

Limitations

- **Language and population.** The study covers English only, and the VAD ratings come from a single, mainly American sample. Emotional vocabulary may be organized differently in other languages and cultures.
- **Word selection.** The 53 words were chosen manually, and five of them had no link at the chosen threshold. A different list, or a lexicon-driven selection, could change the results.
- **Threshold.** Links are defined by a single cosine cut-off (0.50). Testing the robustness of the results across thresholds, or modelling the similarity as a continuous variable, would strengthen the conclusions.
- **Embeddings.** Static Word2Vec vectors conflate senses and place antonyms close together. Contextual embeddings, or a corpus other than news, may give a different picture.
- **Dependence between pairs.** The permutation test addresses this for the distance comparison, but the standard errors of the logistic regression assume independent observations and are likely too small.

5 Conclusion

Comparing distributional and psychological representations of 53 emotion words shows that linguistic usage and affective experience are aligned: semantically related emotions are closer in VAD space. This alignment runs along valence and dominance in roughly equal measure, not along arousal. Emotional vocabulary is therefore organized not only by how good or bad an emotion feels, but also by how much control it grants us.

Code and data. The analysis was carried out in Python (pandas, gensim, SciPy, statsmodels). The notebook, the data and the scripts reproducing every figure and statistic of this report are available at:

https://github.com/NathanTessot75/emotions-language

References

Harris, Z. S. (1954). Distributional structure. *Word*, 10(2–3), 146–162.

Krackhardt, D. (1988). Predicting with networks: Nonparametric multiple regression analysis of dyadic data. *Social Networks*, 10(4), 359–381.

Mehrabian, A., & Russell, J. A. (1974). *An Approach to Environmental Psychology*. MIT Press.

Mikolov, T., Chen, K., Corrado, G., & Dean, J. (2013a). Efficient estimation of word representations in vector space. *arXiv preprint* arXiv:1301.3781. https://arxiv.org/abs/1301.3781

Mikolov, T., Sutskever, I., Chen, K., Corrado, G., & Dean, J. (2013b). Distributed representations of words and phrases and their compositionality. In *Advances in Neural Information Processing Systems 26*.

Osgood, C. E., Suci, G. J., & Tannenbaum, P. H. (1957). *The Measurement of Meaning*. University of Illinois Press.

Russell, J. A. (1980). A circumplex model of affect. *Journal of Personality and Social Psychology*, 39(6), 1161–1178.

Warriner, A. B., Kuperman, V., & Brysbaert, M. (2013). Norms of valence, arousal, and dominance for 13,915 English lemmas. *Behavior Research Methods*, 45(4), 1191–1207. https://doi.org/10.3758/s13428-012-0314-x